UniPlan – Software Requirements Specification (SRS)

Item	Details
Project	UniPlan – Smart Academic Planning & Course Registration System
Document	Software Requirements Specification (SRS) – MVP
Version	1.0 (MVP)
Prepared by	UniPlan Team
Date	9 October 2026

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the **UniPlan MVP**. It is intended for the supervising engineer, developers and testers, and serves as the agreed reference for what the first release must deliver.

1.2 Product Overview

UniPlan is a web-based system that helps university students organize their study plan, see which courses are available to them, register for suitable courses, and track progress toward graduation requirements. It combines four things in one system: the study plan, registration eligibility checking, weekly schedule management, and academic progress tracking.

UniPlan is not only a registration platform. It also tells the student which courses they can register for, which courses are still locked and **why**, and how many credit hours remain available to them in the current semester.

1.3 Objectives

ID	Objective
OBJ-1	Make course planning easier for each student.
OBJ-2	Reduce course registration errors.
OBJ-3	Verify that all registration conditions are met before a registration is confirmed.
OBJ-4	Prevent students from exceeding the maximum allowed credit hours.
OBJ-5	Avoid conflicts between lecture times.

ID	Objective
OBJ-6	Help students track academic progress and the credit hours remaining in their plan.

1.4 Scope

In scope (MVP): two roles only (Student and Admin); authentication and authorization; student academic profile; study plan view; course catalog and search; registration eligibility validation; GPA-based credit-hour limit; course registration and dropping; weekly schedule; academic progress tracking; admin management of courses, plan, sections and registration periods.

Out of scope: see Section 9.

The MVP uses **pre-prepared sample academic data**. No integration with a real university system is required.

1.4.1 Definitions and Abbreviations

Term	Definition
MVP	Minimum Viable Product – the first usable release.
SRS	Software Requirements Specification.
Student	A university student who uses the system to plan and register courses.
Admin	A user who manages courses, the study plan, sections and registration settings.
Course	An academic subject with a code, name and credit hours.
Section	A specific offering of a course in a semester, with a capacity and lecture times.
Prerequisite	A course that must be passed before registering for another course.
Credit Hours	The weight of a course, counted toward registration limits and graduation.
GPA	Cumulative Grade Point Average.
Study Plan	The program's list of courses distributed across semesters.
Registration Period	The time window in which students may register

Term	Definition
	courses.
Drop Period	The time window in which students may drop registered courses.

2. Overall Description

2.1 Product Perspective

UniPlan is a standalone web application with a frontend, a backend that enforces all business rules, and a database. All core validations are executed on the **Backend** to guarantee the correctness of registrations.

2.2 User Classes

Role	Description	Main Capabilities
Student	A university student.	View own academic data and study plan, search courses, register and drop courses, view schedule and progress.
Admin	A system administrator.	Manage courses, study plan, sections, semester and registration/drop periods.

Only these two roles exist in the MVP.

2.3 Operating Environment

The system is a web application accessed through a modern web browser. Planned technology stack: HTML, CSS, JavaScript and Bootstrap (frontend); C# with ASP.NET Core Web API (backend); Entity Framework Core (data access); SQL Server (database); ASP.NET Core Identity for authentication and role-based authorization, with an appropriate mechanism to protect API endpoints.

2.4 Assumptions and Dependencies

- Academic data (students, GPAs, courses, prerequisites, plan, sections, passed courses) is prepared in advance as sample data.
- Credit-hour limit rules are sample, configurable rules and are not universal across universities.
- Each prerequisite is a simple condition (a course that must be passed). Complex or alternative prerequisite groups are not supported.
- One course is registered per operation.

3. Functional Requirements

3.1 Authentication and Authorization

ID	Requirement	Priority
FR-AUTH-01	The system shall allow a student to create an account.	High
FR-AUTH-02	The system shall allow users to log in and log out.	High
FR-AUTH-03	Admin accounts shall be created through a secure method (not through public sign-up).	High
FR-AUTH-04	The system shall assign permissions based on the user's role.	High
FR-AUTH-05	The system shall prevent students from accessing admin pages and operations.	High
FR-AUTH-06	The system shall ensure a student can view only their own data and registrations.	High

3.2 Student Academic Profile

ID	Requirement	Priority
FR-PRO-01	The system shall display the student's basic data.	High
FR-PRO-02	The system shall display the program / study plan the student follows.	High
FR-PRO-03	The system shall display the student's cumulative GPA.	High
FR-PRO-04	The system shall display the courses the student has passed.	High
FR-PRO-05	The system shall display the courses the student is currently registered in.	High

ID	Requirement	Priority
FR-PRO-06	The system shall display the number of credit hours the student has passed.	High
FR-PRO-07	The system shall show a student dashboard summarizing GPA, passed credit hours, hours required by the plan, remaining hours, number of registered courses, credit hours still available to register, and plan progress percentage, with shortcuts to the study plan, course list and schedule.	High

3.3 Study Plan View

ID	Requirement	Priority
FR-PLAN-01	The system shall display the study plan with courses distributed across semesters.	High
FR-PLAN-02	The system shall show the status of each course using the statuses in Section 3.3.1.	High
FR-PLAN-03	When the student selects a Locked course, the system shall display the reason it is not available (for example, a required prerequisite has not been passed).	High
FR-PLAN-04	The plan shall be presented in an organized, simple format (an advanced interactive map is out of scope).	Medium

3.3.1 Course Statuses

Status	Meaning
Completed	The student passed the course.
Registered	The student is currently registered in the course.

Status	Meaning
Available	The course is open for registration and its conditions are met.
Locked	The course is unavailable because a condition is not met.
Not Offered	The course is not offered in the current semester.

3.4 Course Catalog and Search

ID	Requirement	Priority
FR-CAT-01	The system shall display the courses offered in the current semester.	High
FR-CAT-02	The system shall allow searching for a course by name or code.	High
FR-CAT-03	The system shall display each course's name, code and credit hours.	High
FR-CAT-04	The system shall display each course's prerequisites.	High
FR-CAT-05	The system shall display available sections and their lecture times.	High
FR-CAT-06	The system shall display the remaining seats in each section.	High

3.5 Registration Eligibility Validation

Before registering a course, the system shall verify **all** of the following conditions:

ID	Condition checked	Failure behavior
FR-VAL-01	The course is offered in the current semester.	Reject and show reason.
FR-VAL-02	The registration period is open.	Reject and show reason.
FR-VAL-03	The student has passed all prerequisites of the course.	Reject and show reason.

ID	Condition checked	Failure behavior
FR-VAL-04	The student is not already registered in the course.	Reject and show reason.
FR-VAL-05	The selected section has available seats.	Reject and show reason.
FR-VAL-06	Registering the course will not exceed the student's maximum allowed credit hours.	Reject and show reason.
FR-VAL-07	The section's times do not conflict with the student's other registered courses.	Reject and show reason.

ID	Requirement	Priority
FR-VAL-08	If any condition fails, the system shall block the registration and display a message explaining the reason.	High
FR-VAL-09	All core validations shall be executed on the Backend.	High

3.6 Credit-Hour Limit Based on GPA

ID	Requirement	Priority
FR-CRD-01	The system shall determine the student's maximum allowed credit hours based on their cumulative GPA.	High
FR-CRD-02	The system shall calculate the credit hours the student is currently registered in.	High
FR-CRD-03	The system shall calculate the remaining credit hours the student may register.	High
FR-CRD-04	The system shall block any registration that would exceed the allowed limit.	High

ID	Requirement	Priority
FR-CRD-05	The GPA-to-credit-hour rules shall be sample rules that can be modified (configurable, not hard-coded).	High

These rules are examples to demonstrate the idea and are not universal rules for all universities. Initial sample rules (the Admin can modify them):

GPA Range	Max Credit Hours per Semester (sample)
Below 1.50	12
1.50 to 2.00 (inclusive)	15
Above 2.00 up to 3.00 (inclusive)	18
Above 3.00	21

3.7 Course Registration and Dropping

ID	Requirement	Priority
FR-REG-01	The student shall be able to select an available course and section.	High
FR-REG-02	The system shall show the course details, credit hours and lecture times before confirmation.	High
FR-REG-03	The student shall confirm registration only after all conditions in Section 3.5 are verified.	High
FR-REG-04	The student shall be able to view their registered courses.	High
FR-REG-05	The student shall be able to drop a course during the permitted drop period.	High
FR-REG-06	After registration or dropping, the system shall update the registered courses list, credit-hour count and lecture	High

ID	Requirement	Priority
	schedule.	
FR-REG-07	The system shall keep an appropriate record of registration status instead of permanently deleting historical information.	Medium
FR-REG-08	In the MVP, one course shall be registered per operation (bulk registration is out of scope).	High
FR-REG-09	The student shall be able to view a registration history: course, section, semester, registration date, current status, and drop date if dropped.	Medium

3.8 Weekly Schedule

ID	Requirement	Priority
FR-SCH-01	The system shall display the student's weekly schedule based on registered courses and sections.	High
FR-SCH-02	The schedule shall include course name, lecture day, start and end time, and location (if available).	High
FR-SCH-03	The schedule shall update automatically after a course is registered or dropped.	High
FR-SCH-04	The system shall prevent registration in a section that conflicts with the current schedule.	High

3.9 Academic Progress Tracking

ID	Requirement	Priority
FR-PRG-01	The system shall display the	High

ID	Requirement	Priority
	total credit hours required by the study plan.	
FR-PRG-02	The system shall display the credit hours the student has passed.	High
FR-PRG-03	The system shall display the remaining credit hours.	High
FR-PRG-04	The system shall display completed courses and courses not yet passed.	High
FR-PRG-05	The system shall display the progress percentage in the study plan.	High
FR-PRG-06	Earned hours shall be calculated from **passed** courses only, not from currently registered courses.	High

3.10 Admin Management

ID	Area	Requirement	Priority
FR-ADM-01	Courses	Add courses and edit their data.	High
FR-ADM-02	Courses	Set the course code and credit hours.	High
FR-ADM-03	Courses	Define the prerequisites of a course.	High
FR-ADM-04	Study Plan	Define which courses belong to the plan.	High
FR-ADM-05	Study Plan	Distribute courses across semesters.	High
FR-ADM-06	Study Plan	Set the total credit hours required.	High
FR-ADM-07	Sections	Create sections for	High

ID	Area	Requirement	Priority
		offered courses.	
FR-ADM-08	Sections	Set the capacity of each section.	High
FR-ADM-09	Sections	Set lecture times.	High
FR-ADM-10	Sections	Monitor the number of registered students and available seats.	Medium
FR-ADM-11	Semester & Registration	Set the current semester.	High
FR-ADM-12	Semester & Registration	Set the start and end of the registration period.	High
FR-ADM-13	Semester & Registration	Set the permitted course drop period.	High
FR-ADM-14	Courses	Disable a course that is no longer offered; prevent duplicate course codes; keep courses linked to past registrations instead of deleting them.	High
FR-ADM-15	Courses	Validate prerequisites so that a course cannot be a prerequisite of itself.	Medium
FR-ADM-16	Sections	Prevent reducing a section's capacity below its number of registered students, and close a section to new registrations.	High
FR-ADM-17	Credit Rules	Edit the maximum credit hours per GPA range, and ensure ranges do not overlap or leave unintended gaps.	High

ID	Area	Requirement	Priority
FR-ADM-18	Registrations	View the students registered in each section, search a student's registrations, and view the registrations of a given semester.	Medium

Admin changes shall be reflected in the system data.

4. Business Rules

ID	Rule
BR-01	A student cannot register for a course without meeting its prerequisites.
BR-02	The maximum allowed credit hours cannot be exceeded.
BR-03	A student cannot register in a full section.
BR-04	The same course cannot be registered twice in the same semester.
BR-05	A student cannot register in a section whose times conflict with their registered courses.
BR-06	Registration is not allowed outside the defined registration period.
BR-07	A course cannot be dropped outside the permitted drop period.
BR-08	A student cannot access another student's data or modify administrative data.
BR-09	A registered course is not considered completed until it is passed.
BR-10	Registration conditions must be verified on the server before any registration is saved.
BR-11	Section capacity must not be exceeded, even when more than one student tries to register at the same time (concurrency control).
BR-12	Changing a course's or section's data must not

ID	Rule
	corrupt existing registration records.

5. Data Requirements

The system relies on sample data prepared inside the system. No external data collection or university database integration is needed. The sample study plan uses 8 semesters totalling 134 credit hours; this is example data and can be adjusted.

Data Entity	Main Contents
Student accounts	Basic data, login credentials, program, cumulative GPA.
Courses	Code, name, credit hours.
Prerequisites	Relationship between a course and the courses required before it.
Study plan	Courses in the plan, distribution across semesters, total required hours.
Semesters	Current semester, registration period, drop period.
Sections	Course, capacity, lecture days and times, location (optional), registered count.
Student course records	Passed courses, currently registered courses, registration status history.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Passwords shall be stored securely (hashed) and access shall be role-based.
NFR-02	Security	Every protected operation shall check the user's role and ownership of data on the server.
NFR-03	Data Integrity	Registration shall be processed so that section capacity and credit-hour limits are never violated, including concurrent

ID	Category	Requirement
		requests.
NFR-04	Correctness	All eligibility validations shall run on the Backend; frontend checks are for convenience only.
NFR-05	Usability	The interface shall be simple and organized, and every rejection shall show a clear reason.
NFR-06	Maintainability	Credit-hour rules and registration periods shall be configurable without code changes.
NFR-07	Compatibility	The system shall work on modern web browsers.
NFR-08	Extensibility	The design shall allow future additions (see Section 9) without major redesign.

7. External Interface Requirements

- **User interface:** web pages for the Student (profile, plan, catalog, registration, schedule, progress) and for the Admin (management screens).
- **Hardware interfaces:** none.
- **Software interfaces:** none in the MVP; no integration with a university system.
- **Communication:** standard web communication between the browser and the server.

8. MVP Completion (Acceptance) Criteria

The MVP is considered complete when a **student** can do all of the following:

ID	Criterion	Related Requirements
AC-01	Log in and view their academic data.	FR-AUTH-02, FR-PRO-01 to 06
AC-02	View their study plan and the offered courses.	FR-PLAN-01, FR-CAT-01
AC-03	Know which courses are available and why others are	FR-PLAN-02, FR-PLAN-03

ID	Criterion	Related Requirements
	not.	
AC-04	Select a course and section and attempt registration.	FR-REG-01 to 03
AC-05	Receive acceptance or rejection based on clear rules.	FR-VAL-01 to 09, FR-CRD-04
AC-06	See the course in the weekly schedule after successful registration.	FR-SCH-01 to 03
AC-07	Track earned and remaining hours and plan progress.	FR-PRG-01 to 06
AC-08	Drop a course within the permitted period.	FR-REG-05, FR-REG-06

At the same time, the **admin** can manage courses, the study plan, sections and registration periods, and the changes are reflected in the system data.

Final goal: deliver a practical, integrated version proving that UniPlan can organize the study plan, verify registration conditions, and help students track academic progress, leaving additional and advanced features for later phases.

9. Features Outside the MVP

The following are postponed to later phases, to be added if time allows after the MVP is complete:

- Advanced interactive map of course relationships.
- Automatic suggestion of suitable courses for the next semester.
- Generation of a proposed graduation plan for upcoming semesters.
- Complex prerequisite conditions and alternative prerequisite groups.
- Waiting list for full sections.
- Registering multiple courses in one operation.
- Full support for multiple universities and programs.
- Email or SMS notifications.
- Advanced reports and statistics.
- A separate mobile application.
- Integration with the university system to import grades and registrations automatically.

10. Open Points for Engineer Review

These items are not defined in the MVP and need a decision:

#	Point	Note

#	Point	Note
1	API protection mechanism to use with ASP.NET Core Identity.	The stack is defined in Section 2.3; the mechanism is not yet chosen.
2	Final GPA-to-credit-hour rules.	Section 3.6 values are samples only.
3	Password and admin-account creation method details.	Must be secure (FR-AUTH-03).
4	Concurrency approach for section capacity.	Required by BR-11 and NFR-03.